

Introduction to Natural Language Processing

HW 3

Multimodal Retrieval-Augmented Generation

Task introduction

- Building your own Retrieval-Augmented Generation (RAG) retriever that, given a question about a document, selects the most relevant pieces of evidence from the document to support answering that question.
- The documents in this task are multimodal, containing both text and images. We provide textual descriptions for every image, so a text-only retriever is sufficient; using the raw images (e.g., via a VLM) is optional.
- For each question, return a single ranked list of the top-5 most relevant evidence items overall (text and images combined, not 5 per modality), ordered from most to least relevant. Your goal is to maximize the overlap with the ground-truth evidence.
- Note: This assignment only evaluates the retrieval stage — you do not need to generate final answers.
- **Requirement :**
 - Submit a report and your source code to E3
 - Upload your submission to Kaggle

Task rules & constraints

- **Model size limit:** Your model must be $\leq 80\text{B}$ total parameters (MoE included — count total params, not active). We judge this by the model's advertised size in its official name: anything labeled 80B or below qualifies, regardless of the exact parameter count. **Open-weight** only (e.g., Qwen3-80B-A3B, Llama-3.1-70B, Gemma-3-27B).
- **Any approach is allowed:** You are free to use any approach you like — the only constraints are **the model size limit and the open-weight requirement**.
- **API usage policy:** Any API used in your retrieval pipeline (embedding, reranking, query rewriting, etc.) must be backed by an **open-weight model within the 80B parameter limit**. **Closed-source APIs such as GPT-5**, Claude are **NOT** allowed.

Dataset

The dataset consists of question-document pairs built from real-world long documents, where relevant evidence may appear in different modalities such as text, tables, and figures.

- HW3_dataset

File	Data_nums	Description
train.jsonl	2055	Labeled samples for training and experimentation
test.jsonl	1798	Unlabeled samples for submission
image	14826	Image assets referenced by the samples
sample_submission.csv	1798	Example submission file

Data Introduction

Each sample in dev_15.jsonl is a JSON object:

- q_id: Unique question ID.
- doc_name / domain: Source document name and its category.
- question: Natural-language question to answer from the document.
- evidence_modality_type: Modalities that may contain evidence, such as text, table, figure, or chart.
- text_quotes: A list of candidate text snippets, each with a quote_id and the original text content..
- img_quotes: A list of candidate image evidences (tables / charts / figures), each with a quote_id, an img_path pointing to the file under image/, and an img_description a pre-generated natural-language description of the image. You may use this description directly as text evidence, or generate your own with a multimodal model if you prefer.
- gold_quotes: Ground-truth supporting quote_ids. Only available in dev_15.jsonl.
- answer_short / answer_interleaved: Reference answers for development only.
- **Note:** test.jsonl has the same public fields but excludes gold labels and reference answers.

Kaggle Submission (70%)

- [Kaggle link](#)

Use your [@nycu.edu.tw](#) email address to join the competition.

- Display team name: <student ID>

- team name error: -5%

- Submission format

- A CSV file with 2 columns: q_id, gold_quotes.
 - Each row corresponds to one question in test.jsonl.
 - q_id must match the question ID exactly.
 - gold_quotes contains your top-5 retrieved quote_ids, separated by a single space.
 - Column names must be exactly q_id, gold_quotes.

- There are one simple baseline and one strong baseline, beat them to get the higher score.

- Evaluation metric: Recall@5. Submit at most 5 quote_ids per row; extra ones beyond the 5th will be ignored. Submitting fewer than 5 is allowed but reduces your maximum recall.

q_id	gold_quotes
0	text1 text2 text3 text4 image5

Change your team name

Please remember to change the team name to <student ID>, or there will be a deduction of 5 points for HW 3.

2026 INLP HW3

Multimodal Retrieval-Augmented Generation



Settings Overview Data Code Models Discussion Leaderboard Rules **Team**



You're off to a great start! You've completed 8 of 11 tasks to launch your competition.

Launch Checklist

Your Team

Everyone that competes in a Competition does so as a team - even if you're competing by yourself. [Learn more.](#)

General



TEAM NAME

yourStudentID

This name will appear on your team's leaderboard position.

Kaggle Submission (70%)

- Public leaderboard is calculated with approximately 30% of the test data, private leaderboard is calculated with the other 70%, so the final standings may be different.
You can only view your private leaderboard score after the competition has ended.
- Your submission is evaluated by **Recall@5**, which measures how many ground-truth evidence quote_ids are covered by your top-5 retrieved results.
- You can submit **at most 5 times each day** and choose 3 of the submissions to be considered for the private leaderboard, or will otherwise default to the best public scoring submissions.

Report Submission

Please answer the following 4 questions in detail:

Q1. Method Description (5%)

- Clearly describe your overall retrieval pipeline and the main idea behind your method.
- Explain the retriever, evidence preprocessing, ranking procedure, and any additional techniques you used.

Q2. Comparison of Retrieval Methods (5%)

- Based on your experimental results, compare the performance of BM25, Dense Retriever, direct LLM selection, and your own method.
- Analyze the strengths and weaknesses of all four methods, and explain what you think causes the performance differences.

Report Submission

Q3. Multimodal Embedding vs. Text-Description Retrieval for Image Evidence(10%)

There are two common ways to retrieve image evidence :

- (a) Use a multimodal embedding model to encode the image directly into a vector, then compare it with the question embedding.
- (b) Convert each image into a text description, then retrieve it together with other text evidence using a text-only retriever. You may either use the `img_description` field that is already provided in the dataset, or generate your own description with a VLM / image-captioning model.
- Briefly state which option you chose and why.
- Compare your experimental results for these two approaches. Discuss which one performs better on this task and explain what you think causes the difference.

Q4. Modality Preference Analysis (10%)

Examine whether your retriever shows a preference for a particular modality in its retrieval results.

- For example, does it tend to retrieve more text evidence, or does it favor image evidence?
- Based on your actual experimental results, describe this preference and analyze what you think causes it.
- even if your retriever shows no clear preference, you are still expected to explain what aspects of your method cause this balanced behavior.

Please answer the questions in detail to get full credit for each question

Grading policy

- Kaggle (70%)
 - 30% based on the public leaderboard score and 70% based on the private leaderboard score
 - Basic score :
 - Over strong baseline : 55
 - Over simple bassline : 40
 - Under simple baseline : 25
 - Ranking score:
 $15 - (15/N) * (\text{ranking} - 1)$, N=numbers of people in the interval
- Report (30%)
 - 10% for each quesiton

You will receive 0 points if you do not submit the source code.

E3 Submission

Submit your source code and report to E3 before 5/26(Tue) 23:59, **no late submissions will be accepted !**

Format:

- HW3_<student ID>.zip
 - Source code : HW3_<student ID>.py or HW3_<student ID>.ipynb
 - report : HW3_<student ID>.pdf

If you have any question about HW 3, please feel free to contact with TA: Ting-Shian Kuo through email chad910713.cs14@nycu.edu.tw

Have Fun !

